

# ALEXEI OBLOMKOV

Professor of Mathematics

Department of Mathematics and Statistics, University of Massachusetts Amherst

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Citizenship: Russia; US Permanent Resident since 2010; US citizen since 2014.

## RESEARCH INTERESTS

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**Representation theory:** double affine Hecke algebras, symplectic reflection algebras, Cherednik algebras, special functions and orthogonal polynomials. **Algebraic geometry:** Hilbert schemes, Gromov–Witten and Donaldson–Thomas invariants, affine Springer fibers, noncommutative and projective geometry. **Low-dimensional topology:** knot homology, HOMFLYPT homology, categorification. **Mathematical physics:** integrable systems, inverse scattering, topological quantum field theory.

## EDUCATION

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|-----------|-----------------------------------------------------------------------------------------------|
| 2005      | <b>Ph.D. in Mathematics</b><br>Massachusetts Institute of Technology. Advisor: Pavel Etingof. |
| 1999–2001 | <b>Graduate study</b><br>Independent University of Moscow; Moscow State University.           |
| 1998      | <b>M.A. in Pure Mathematics</b><br>Independent University of Moscow; Moscow State University. |

## EMPLOYMENT

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| 2020–         | <b>Professor</b><br>University of Massachusetts, Amherst.                               |
| 2018 (Spring) | <b>Research Professor</b><br>Mathematical Sciences Research Institute (MSRI), Berkeley. |
| 2015–2019     | <b>Associate Professor</b><br>University of Massachusetts, Amherst.                     |
| 2009–2015     | <b>Assistant Professor</b><br>University of Massachusetts, Amherst.                     |
| 2008–2009     | <b>Visiting Assistant Professor</b><br>Princeton University.                            |
| 2006–2008     | <b>Instructor</b><br>Princeton University.                                              |
| 2005–2006     | <b>Postdoctoral Fellow</b><br>Institute for Advanced Study, Princeton.                  |

## GRANTS

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|           |                                                          |
|-----------|----------------------------------------------------------|
| 2022–2025 | NSF DMS-2200798, \$217,038, sole PI.                     |
| 2018–2021 | NSF FRG DMS-1760373, \$300,000 for UMass.                |
| 2015–2016 | NSF DMS-1546311, workshop funding, \$25,000, sole co-PI. |

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|-----------|---------------------------------------------|
| 2014–2019 | NSF CAREER DMS-1352398, \$420,000, sole PI. |
| 2010–2013 | NSF DMS-1001609, \$123,910, sole PI.        |
| 2007–2010 | NSF DMS-0701367, \$111,300, sole PI.        |

#### AWARDS AND HONORS

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|            |                                                                                                                               |
|------------|-------------------------------------------------------------------------------------------------------------------------------|
| 2018       | Simons Fellow in Mathematics, Fall 2018.                                                                                      |
| 2014–2019  | NSF CAREER Award.                                                                                                             |
| 2010–2012  | Alfred P. Sloan Research Fellowship.                                                                                          |
| 2004       | Charles and Jennifer Johnson Prize, MIT departmental award for the best paper submitted by a graduate student.                |
| 2003, 2004 | Rogers Family Prize, MIT departmental award for the best mentor–mentee team in the Summer Program for Undergraduate Research. |

#### PUBLICATIONS

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##### Preprints and submitted work

15. *Asymptotic  $q, t$ -Fuss–Catalan numbers for type  $B$* , arXiv:2609.11691.
14. (with Y. Huang, R. Jiang) *Quot scheme of points on torus knot singularities*, arXiv:2608.16086.
13. (with I. Cavey, E. Gorsky, J. Turner) *Wall-crossing for Hilbert schemes*, arXiv:2607.05624.
12. (with L. Rozansky) *A categorification of representations of  $U_q(\mathfrak{gl}_{1|1})$* , arXiv:2510.13039.
11. (with I. Cavey, E. Gorsky, J. Turner) *Newton–Okounkov bodies for nested Hilbert schemes*, arXiv:2510.07420.
10. (with O. Kivinen, G. Wyss) *Discriminants and motivic integration*, arXiv:2503.18449.
9. (with L. Kamenova, G. Mongardi) *Fixed loci of symplectic automorphisms of  $K3^{[n]}$  and  $n$ -Kummer type manifolds*, arXiv:2308.14692.
8. (with L. Rozansky) *Matrix factorizations and  $gl(m|k)$ -quantum invariants*, arXiv:2212.02665.
7. (with L. Rozansky) *Soergel bimodules and matrix factorizations*, arXiv:2010.14546.
6. (with N. Bottman) *A compactification of the moduli space of marked vertical lines in  $\mathbb{C}^2$* , arXiv:1910.02037.
5. (with L. Rozansky) *Dualizable link homology*, arXiv:1905.06511.
4. *EGL formula for DT/PT theory of local curves*, arXiv:1901.03014.
3. (with E. Carlsson) *Affine Schubert calculus and double coinvariants*, arXiv:1801.09033.
2. (with L. Rozansky) *A categorification of a cyclotomic Hecke algebra*, arXiv:1801.06201.
1. (with Z. Yun) *The cohomology ring of certain compactified Jacobians*, arXiv:1710.05391.

##### Published and accepted papers

41. (with E. Gorsky, M. Mazin) *Generic curves and non-coprime Catalans*, Algebraic Geometry (2026), arXiv:2210.12569.
40. (with E. Gorsky, O. Kivinen) *The affine Springer fiber – sheaf correspondence*, Advances in Mathematics **464** (2025), arXiv:2204.00303.
39. (with L. Rozansky) *Categorical Chern character and braid groups*, Adv. Math. **437** (2024), 66 pp., arXiv:1811.03257.

38. (with L. Rozansky) *3D TQFT and HOMFLYPT homology*, Lett. Math. Phys. **113** (2023), no. 3, Paper No. 71, 62 pp., arXiv:1812.06340.
37. (with L. Rozansky) *HOMFLYPT homology of Coxeter links*, Transform. Groups **28** (2023), no. 3, 1245–1275, arXiv:1706.00124.
36. (with M. Moreira, A. Okounkov, R. Pandharipande) *Virasoro constraints for stable pairs on toric threefolds*, Forum Math. Pi **10** (2022), Paper No. e20, 62 pp., arXiv:2008.12514.
35. (with L. Kamenova, G. Mongardi) *Symplectic involutions of  $K3^{[n]}$  type and Kummer  $n$  type manifolds*, Bull. Lond. Math. Soc. **54** (2022), no. 3, 894–909, arXiv:1809.02810.
34. (with A. Okounkov, R. Pandharipande) *GW/PT descendent correspondence via vertex operators*, Comm. Math. Phys. **374** (2020), no. 3, 1321–1359, arXiv:1806.00714.
33. (with L. Rozansky) *Affine braid group, JM elements and knot homology*, Transform. Groups **24** (2019), no. 2, 531–544, arXiv:1702.03569.
32. *Notes on matrix factorizations and knot homology*, Lecture Notes in Math. **2248**, Fond. CIME/CIME Found. Subser., Springer, Cham, 2019, 83–127, arXiv:1901.04052.
31. (with L. Rozansky) *Knot homology and sheaves on the Hilbert scheme of points on the plane*, Selecta Math. (N.S.) **24** (2018), no. 3, 2351–2454, arXiv:1608.03227.
30. (with V. Shende, J. Rasmussen) *The Hilbert scheme of a plane curve singularity and HOMFLY homology of its link*, Geom. Topol. **22** (2018), no. 2, 645–691, arXiv:1201.2115.
29. (with Z. Yun) *Geometric representations of graded and rational Cherednik algebras*, Adv. Math. **292** (2016), 601–706, arXiv:1407.5685.
28. (with S. Nawata) *Lectures on knot homology*, Proceedings of the Workshop on Physics and Mathematics of Link Homology, Centre de Recherches Mathématiques, Université de Montréal, 137–177, Contemp. Math. **680**, Amer. Math. Soc., Providence, RI, 2016, arXiv:1510.01795.
27. (with E. Gorsky, V. Shende, J. Rasmussen) *Torus knots and rational DAHA*, Duke Math. J. **163** (2014), no. 15, 325–401, arXiv:1207.4523.
26. (with E. Gorsky, J. Rasmussen) *On stable Khovanov homology of torus knots*, Exp. Math. **22** (2013), no. 3, 265–281, arXiv:1206.2226.
25. (with V. Shende) *The Hilbert scheme of a plane curve singularity and the HOMFLY polynomial of its link*, Duke Math. J. **161** (2012), no. 7, 1271–1303, arXiv:1003.1568.
24. (with D. Maulik, A. Okounkov, R. Pandharipande) *The Gromov–Witten/Donaldson–Thomas correspondence for toric 3-folds*, Invent. Math. **186** (2011), no. 2, 435–479, arXiv:0809.3976.
23. (with E. Stoica) *Finite dimensional representations of the double affine Hecke algebra of rank 1*, J. Pure Appl. Algebra **213** (2009), no. 5, 766–771, arXiv:math.RT/0409256.
22. (with D. Maulik) *Quantum cohomology of the Hilbert scheme of points on  $A_n$ -resolutions*, J. Amer. Math. Soc. **22** (2009), 1055–1091, arXiv:0802.2737.
21. (with D. Maulik) *Donaldson–Thomas theory of  $A_n \times \mathbb{P}^1$* , Compos. Math. **145** (2009), no. 5, 1249–1276, arXiv:0802.2739.
20. (with P. Etingof, S. Loktev, L. Rybnikov) *A Lie-theoretic construction of spherical symplectic reflection algebras*, Transform. Groups **13** (2008), no. 3–4, 541–556, arXiv:0801.2339.
19. (with P. Etingof, W. L. Gan, V. Ginzburg) *Harish-Chandra homomorphisms and symplectic reflection algebras for wreath-products*, Publ. Math. IHÉS **105** (2007), 91–155, arXiv:math.RT/0511489.
18. *Deformed Harish-Chandra homomorphism for the cyclic quiver*, Math. Res. Lett. **33** (2007), no. 3,

- 359–372, arXiv:math.RT/0504395.
17. (with P. Etingof, E. Rains) *Generalized double affine Hecke algebras of rank 1 and quantized Del Pezzo surfaces*, Adv. Math. **212** (2007), no. 10, 749–796, arXiv:math.QA/0406480.
  16. (with P. Etingof, W. L. Gan) *Generalized double affine Hecke algebras of higher rank*, J. Reine Angew. Math. (Crelle) **600** (2006), 177–201, arXiv:math.QA/0504089.
  15. (with P. Etingof) *Quantization, orbifold cohomology, and Cherednik algebras*, Proceedings of the JHLM Workshop, Contemp. Math. **417** (2006), 171–182, arXiv:math.QA/0311005.
  14. (with A. Penskoi) *Laplace transformations and spectral theory of two-dimensional semi-discrete and discrete hyperbolic Schrödinger operators*, IMRN (2005), no. 18, 1089–1126, arXiv:math-ph/0311036.
  13. (with J. V. Stokman) *Vector valued spherical functions and Macdonald–Koornwinder polynomials*, Compositio Math. **141** (2005), no. 5, 1310–1350, arXiv:math.QA/0311512.
  12. *Heckman–Opdam’s Jacobi polynomials for the  $BC_n$  root system and generalized spherical functions*, Adv. Math. **186** (2004), no. 1, 153–180, arXiv:math.RT/0202076.
  11. *The double affine Hecke algebras and Calogero–Moser spaces*, Represent. Theory **8** (2004), 243–266, arXiv:math.RT/0303190.
  10. *Double affine Hecke algebras of rank 1 and affine cubic surfaces*, IMRN (2004), no. 18, 877–912, arXiv:math.RT/0306393.
  9. (with O. Chalykh, P. Etingof) *Generalized Lamé operators*, Comm. Math. Phys. **239** (2003), no. 1–2, 115–153, arXiv:math.QA/0212029.
  8. *The isoenergy spectral problem for multidimensional difference operators*, (Russian) Funktsional. Anal. i Prilozhen. **36** (2002), no. 2, 45–61; translation in Funct. Anal. Appl. **36** (2002), no. 2, 120–133.
  7. *On the spectral properties of two classes of difference periodic operators*, (Russian) Mat. Sb. **193** (2002), no. 4, 87–112; translation in Sb. Math. **193** (2002), no. 3–4, 559–584.
  6. *Difference operators on two-dimensional regular lattices*, (Russian) Teoret. Mat. Fiz. **127** (2001), no. 1, 34–46; translation in Theoret. and Math. Phys. **127** (2001), no. 1, 435–445.
  5. (with A. Penskoi) *Two-dimensional algebro-geometric difference operators*, J. Phys. A **33** (2000), no. 50, 9255–9264, arXiv:math-ph/0010024.
  4. (with O. Chalykh) *Harmonic oscillator and Darboux transformations in many dimensions*, Phys. Lett. A **267** (2000), no. 4, 256–264.
  3. (with F. L. Zak, A. V. Inshakov, S. M. L’vovski) *On congruences of lines of order one in  $\mathbb{P}^3$* , preprint (1999), available at <http://www.mccme.ru/iium/postscript/s99/notes>.
  2. *Monodromy-free Schrödinger operators with quadratically increasing potential*, (Russian) Teoret. Mat. Fiz. **121** (1999), no. 3, 374–386; translation in Theoret. and Math. Phys. **121** (1999), no. 3, 1574–1584.
  1. *Integrability of some quantum systems associated with the root system  $B_2$* , (Russian) Vestnik Moskov. Univ. Ser. I Mat. Mekh. **72** (1999), no. 2, 6–9; translation in Moscow Univ. Math. Bull. **54** (1999), no. 2, 5–8.

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## INVITED TALKS AND LECTURES

### 2024

- CalTech, Algebraic Geometry Seminar, 10/2024.
- EPFL, Switzerland, week-long lecture course on TQFT, 10/2024.
- UC Davis, Geometry Seminar, 06/2024.

- *Les Diablerets, Switzerland*, workshop on categorical topological invariants, 01/2024.

**2023**

- *MIT*, Representation Theory Seminar, 11/2023.
- *Johns Hopkins University*, Topology Seminar, 10/2023.
- *University of Notre Dame*, workshop on matrix factorizations, 10/2023.
- *Simons Center for Geometry and Physics*, lectures on the GW/PT correspondence, 09/2023.
- *American Institute of Mathematics*, workshop on knot homology, 08/2023.
- *UC Davis*, Algebraic Geometry Seminar, 06/2023.

**2022**

- *Stony Brook University*, Algebraic Geometry Seminar, 12/2022.
- *Yale University*, Representation Theory Seminar, 11/2022.
- *Harvard University*, Mathematical Physics Seminar, 10/2022.
- *Washington University*, Geometry Seminar, 05/2022.

**2021**

- *University of Hamburg*, Mathematical Physics Colloquium, 10/2021.

**2019**

- *University of Edinburgh, UK*, Geometric Representation Theory and Low-Dimensional Topology, 06/2019.
- *Schloss Hünigen, Switzerland*, Helvetic Algebraic Geometry Seminar, 06/2019.
- *ETH Zürich, Switzerland*, Algebraic Geometry Seminar, 06/2019.
- *CalTech*, Hidden Algebraic Structures in Topology, 04/2019.

**2018**

- *Columbia University*, Informal Mathematical Physics Seminar, 10/2018.
- *ETH Zürich, Switzerland*, Algebraic Geometry Seminar, 10/2018.
- *MSRI*, Enumerative Geometry Seminar, 04/2018.
- *UC Davis*, Geometry and Topology Seminar, 04/2018.
- *CIME, Cetraro, Italy*, three lectures at the Summer School on Geometric Representation Theory and Gauge Theory, 2018.

**2017**

- *Perimeter Institute, Canada*, Mathematical Physics Seminar, 10/2017.
- *University of Waterloo, Canada*, Colloquium, 10/2017.
- *University of Hamburg, Germany*, String Math, invited speaker, 07/2017.
- *CalTech*, Mathematical Physics Seminar, 05/2017.
- *MIT*, Representation Theory Seminar, 03/2017.

**2016**

- *Fields Institute, Canada*, one-week workshop on Hall algebras, enumerative invariants and gauge theory, invited speaker, 11/2016.
- *ICTP, Trieste, Italy*, one-week workshop on gauge theory and related fields, invited speaker, 09/2016.
- *CRM, Montreal, Canada*, one-week workshop on algebraic cycles, invited speaker, 06/2016.

- *Northeastern University*, Algebraic Geometry Seminar, 03/2016.

## 2015

- *Texas A&M University*, Geometry and Combinatorics Seminar, 10/2015.
- *IAS/Park City Mathematics Institute*, Summer School on Geometric Representation Theory, invited speaker, 06/2015.
- *Stony Brook University*, one-week workshop on knot homology at SCGP, invited speaker, 05/2015.

## 2014

- *CRM, Montreal, Canada*, two-week workshop, four lectures on knot homology, DAHA and singular curves, 07/2014.
- *Banff International Research Station*, one-week workshop on topological recursion, quantum curves and quantum invariants, invited speaker, 06/2014.
- *Northwestern University*, conference on QFT and representation theory, invited speaker, 05/2014.
- *University of Geneva, Switzerland*, conference on quantum invariants and geometry, invited speaker, 04/2014.
- *EPFL, Switzerland*, three lectures, Geometry Seminar, 04/2014.
- *University of Maryland*, Colloquium, 04/2014.
- *Northwestern University*, Topology Seminar, 03/2014.
- *UC Davis*, three lectures on knot invariants and algebraic geometry, 02/2014.

## 2013

- *Stanford University*, Algebraic Geometry Seminar, 09/2013.
- *Banff International Research Station*, one-week workshop on motivic DT invariants, invited speaker, 06/2013.
- *MSRI*, conference on noncommutative geometry, invited speaker, 05/2013.
- *University of Michigan, Ann Arbor*, Lie Theory Seminar, 04/2013.

## 2011

- *Rutgers University*, Mathematical Physics Seminar, 12/2011.
- *University of North Carolina, Chapel Hill*, Geometry and Mathematical Physics Seminar, 12/2011.
- *University of North Carolina, Chapel Hill*, Colloquium, 12/2011.
- *Oxford University, UK*, series of lectures on knot invariants, the Hilbert scheme of points on curves and DAHAs, 11/2011.
- *Simons Center for Geometry and Physics*, conference on homological invariants in low-dimensional topology, 06/2011.
- *Simons Center for Geometry and Physics*, conference on mirror symmetry and equivariant cohomology, 05/2011.
- *University of Texas, Austin*, Geometry Seminar, 03/2011.
- *University of Michigan, Ann Arbor*, Mathematical Physics Seminar, 02/2011.

## 2010

- *University of Illinois at Urbana–Champaign*, Wall-Crossing in Mathematics and Physics, 05/2010.
- *University of Illinois, Chicago*, Algebraic Geometry Seminar, 03/2010.

## 2009

- *University of Wisconsin, Madison*, Geometry Seminar, 11/2009.
- *University of Chicago*, Representation Theory Seminar, 11/2009.
- *University of Michigan, Ann Arbor*, Mathematical Physics Seminar, 10/2009.
- *Boston University*, Geometry Seminar, 10/2009.
- *MIT*, Noncommutative Algebra Seminar, 05/2009.
- *University of Illinois at Urbana–Champaign*, AMS Special Session on TQFT, 04/2009.
- *Cornell University*, Representation Theory Seminar, 04/2009.
- *Stony Brook University*, Colloquium, 01/2009.

## 2008

- *University of Massachusetts, Amherst*, special seminar, 12/2008.
- *University of Texas, Austin*, special seminar, 12/2008.
- *University of Illinois at Urbana–Champaign*, Algebraic Geometry Seminar, 12/2008.
- *University of Michigan*, special seminar, 11/2008.

## 2002–2007

- *Northwestern University*, Mathematical Physics Seminar, 10/2007.
- *University of California, Riverside*, Lie Theory Seminar, 12/2006.
- *University of Virginia, Charlottesville*, Colloquium, 11/2006.
- *University of Virginia, Charlottesville*, Algebra Seminar, 11/2006.
- *University of Massachusetts, Amherst*, Representation Theory Seminar, 12/2005.
- *University of Oregon*, AMS Special Session on Noncommutative Algebra and Noncommutative Birational Geometry, 11/2005.
- *University of Washington*, Algebra Seminar, 11/2005.
- *Rutgers University*, Lie Group Seminar, 10/2005.
- *UC Santa Barbara*, AMS Special Session on Algebraic Geometry and Combinatorics, 04/2005.
- *University of Michigan*, Algebra Seminar, 12/2004.
- *MIT*, Infinite-Dimensional Algebra Seminar, 04/2004.
- *Yale University*, Geometry, Symmetry and Physics, 12/2003.
- *University of Massachusetts, Amherst*, Representation Theory Seminar, 12/2003.
- *Cornell University*, Lie Groups Seminar, 01/2003.
- *MIT*, Infinite-Dimensional Algebra Seminar, 11/2002.
- *University of Amsterdam*, lectures on special functions, 07/2002.

## EDITORIAL SERVICE

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2025–

### **Managing Editor, *Galileo***

Mathematical Sciences Publishers (MSP). *Galileo* publishes research and research-expository work across the mathematical sciences, with an emphasis on new phenomena and on conveying the discovery process; first issue 2027.

### **Refereeing for journals**

Annals of Mathematics; Inventiones Mathematicae; Journal of the AMS; Communications in Mathematical Physics; Advances in Mathematics; Compositio Mathematica; Selecta Mathematica; IMRN; Transactions of

the AMS; Transformation Groups; Journal of Algebra; Journal of Algebraic Combinatorics A; Contemporary Mathematics.

## PROFESSIONAL SERVICE

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### Grant proposal review

NSERC (Canada); NSA; Swiss National Science Foundation; the Netherlands Organisation for Scientific Research (NWO); the Chilean National Science and Technology Commission; Newton Institute scientific proposals. Served on three NSF panels.

### Conference organization

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|-----------|--------------------------------------------------------------------------|
| 2025      | Co-organizer, month-long program, Bernoulli Center, EPFL, Lausanne.      |
| 2023      | Co-organizer, two-month program, Simons Center for Geometry and Physics. |
| 2023      | Co-organizer, AIM workshop.                                              |
| 2021–2022 | Co-organizer, AIM research community.                                    |
| 2022      | Co-organizer, one-week FRG workshop, University of Oregon.               |
| 2020      | Co-organizer, one-week FRG workshop, UMass Amherst.                      |
| 2019      | Co-organizer, one-week FRG workshop, UC Davis.                           |
| 2018      | Co-organizer, one-week AIM workshop.                                     |
| 2017      | Co-organizer, three-day workshop, Amherst.                               |
| 2015      | Co-organizer, three-day workshop, Amherst.                               |
| 2014      | Co-organizer, one-week workshop, Banff.                                  |

## DEPARTMENTAL AND UNIVERSITY SERVICE

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### University of Massachusetts, Amherst

- Co-organizer, Valley Geometry Seminar and Representation Theory Seminar (2009–); Quantum Field Theory Seminar (Fall 2010).
- Lectures in the Algebraic Geometry Reading Seminar (Spring 2009, 5 lectures; Spring 2010, 2; Fall 2010, 2; Spring 2011, 2; Spring 2012, 2).
- Colloquium Committee (Fall 2010, Fall 2013, Spring 2014, 2016–2017).
- Advanced Graduate Algebra Exam Committee (Fall 2011–Spring 2013); Advanced Graduate Topology Exam Committee (Fall 2014–Spring 2023).
- Faculty Search Committee (Fall 2011, Spring 2012, Fall 2012, Spring 2013, Spring 2019); Chair, Search Committee for Visiting Assistant Professor (2018–).
- Undergraduate Advisor (2010–); Undergraduate Affairs Committee (Fall 2017); Climate Committee (2023).
- Trained the UMass Putnam team (2015, 2016, 2022).

### Princeton University

- Co-organizer, Princeton Algebraic Geometry Seminar (Spring 2009).
- Served on six general exam committees; advised an undergraduate senior thesis.

PH.D. ADVISING

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| 2024 | <b>Arthur Wang</b><br><i>Khovanov homology and matrix factorizations</i> , UMass Amherst.                                                                                                              |
| 2016 | <b>Thomas Shelly</b><br><i>Skein theory and algebraic geometry for the two-variable Kauffman invariant of links</i> , UMass Amherst. Subsequently Visiting Assistant Professor, Mount Holyoke College. |
| 2015 | <b>Tobias Wilson</b><br><i>Topology of the affine Springer fiber in type A</i> , UMass Amherst. Subsequently at Bloomberg.                                                                             |

**Current students**

- Pranav Kalkunte — motivic integration.
- Chuijiao Zhang — Springer theory for DAHAs.

UNDERGRADUATE RESEARCH MENTORING

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|-------------|----------------------------------------------------------------|
| Summer 2025 | REU program, UMass Amherst — mentored a team of four students. |
| Summer 2019 | REU program, UMass Amherst — team of four students.            |
| Summer 2018 | REU program, UMass Amherst — team of four students.            |
| Summer 2017 | REU program, UMass Amherst — team of four students.            |
| Summer 2016 | REU program, UMass Amherst — team of four students.            |
| Summer 2015 | REU program, UMass Amherst — team of four students.            |
| Summer 2013 | REU program, UMass Amherst — one student.                      |
| Summer 2004 | Summer Program in Undergraduate Research (SPUR), MIT.          |
| Summer 2003 | Summer Program in Undergraduate Research (SPUR), MIT.          |

TEACHING

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**University of Massachusetts, Amherst**

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|-------------|----------------------------------------------|
| Fall 2025   | Introduction to Lie Theory (Math 718).       |
| Spring 2025 | Graduate Algebra (Math 613).                 |
| Fall 2024   | Graduate Algebra (Math 612).                 |
| Spring 2024 | Graduate Complex Analysis (Math 621).        |
| Spring 2024 | Introduction to Linear Algebra (Math 235).   |
| Fall 2023   | Introduction to Complex Analysis (Math 421). |
| Spring 2023 | Introduction to Linear Algebra (Math 235).   |
| Fall 2022   | Representation Theory (Math 680).            |
| Fall 2022   | Putnam Seminar (Math 490).                   |
| Spring 2022 | Introduction to Complex Analysis (Math 421). |
| Fall 2021   | Multivariable Calculus (Math 233).           |
| Spring 2021 | Graduate Topology II (Math 672).             |
| Fall 2020   | Multivariable Calculus (Math 233).           |
| Fall 2020   | Graduate Topology I (Math 671).              |

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|-------------|----------------------------------------------------------------------------------------|
| Spring 2020 | Introduction to Linear Algebra (Math 235).                                             |
| Fall 2019   | Introduction to Linear Algebra (Math 235).                                             |
| Fall 2019   | Graduate Topology I (Math 671).                                                        |
| Spring 2019 | Graduate Topology II (Math 672).                                                       |
| Spring 2019 | Advanced Multivariable Calculus (Math 425).                                            |
| Fall 2017   | Introduction to Linear Algebra (Math 235).                                             |
| Fall 2016   | Introduction to Topology (Math 671).                                                   |
| Fall 2016   | Introduction to Linear Algebra (Math 235).                                             |
| Fall 2015   | Introduction to Linear Algebra (Math 235).                                             |
| Fall 2015   | Introduction to Representation Theory (Math 797).                                      |
| Spring 2015 | Introduction to Linear Algebra (Math 235) — course chair for five additional sections. |
| Fall 2014   | Introduction to Topology (Math 671).                                                   |
| Fall 2014   | Introduction to Linear Algebra, Honors (Math 235H).                                    |
| Fall 2013   | Knot Invariants (Math 797KI).                                                          |
| Fall 2013   | Introduction to Linear Algebra (Math 235).                                             |
| Fall 2012   | Putnam Exam (Math 491A).                                                               |
| Fall 2012   | Multivariable Calculus (Math 233).                                                     |
| Fall 2012   | Introduction to Linear Algebra (Math 235).                                             |
| Fall 2011   | Introduction to Linear Algebra (Math 235).                                             |
| Fall 2010   | Graduate Algebra (Math 631).                                                           |
| Spring 2010 | Calculus I (Math 131, 2 sections) — lectures and problem sessions.                     |

### **Princeton University**

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|-------------|----------------------------------------------------------------------------------------|
| Spring 2009 | Multivariable Calculus (MAT 201) — review sessions.                                    |
| Spring 2009 | Junior Seminar: Representation Theory of Finite Groups — organized the seminar.        |
| Fall 2008   | Advanced Multivariable Calculus (MAT 203), course head — lectures and review sessions. |
| Spring 2008 | Multivariable Calculus (MAT 201, 2 sections) — lectures and problem sessions.          |
| Fall 2007   | Advanced Multivariable Calculus (MAT 203, 2 sections) — lectures and problem sessions. |
| Spring 2007 | Multivariable Calculus (MAT 201, 2 sections) — lectures and problem sessions.          |
| Fall 2006   | Advanced Multivariable Calculus (MAT 203, 2 sections) — lectures and problem sessions. |

### **Massachusetts Institute of Technology**

|             |                                                                                   |
|-------------|-----------------------------------------------------------------------------------|
| Spring 2004 | Linear Algebra (18.06) — problem sessions.                                        |
| Fall 2003   | Advanced Calculus (18.01A) and Multivariable Calculus (18.02) — problem sessions. |
| Spring 2003 | Differential Equations (18.03) — problem sessions and grading.                    |
| 2001–2002   | Grading for two graduate-level courses.                                           |

### **Independent University of Moscow**

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| 1999–2001 | Advanced algebra courses — problem sessions. |
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